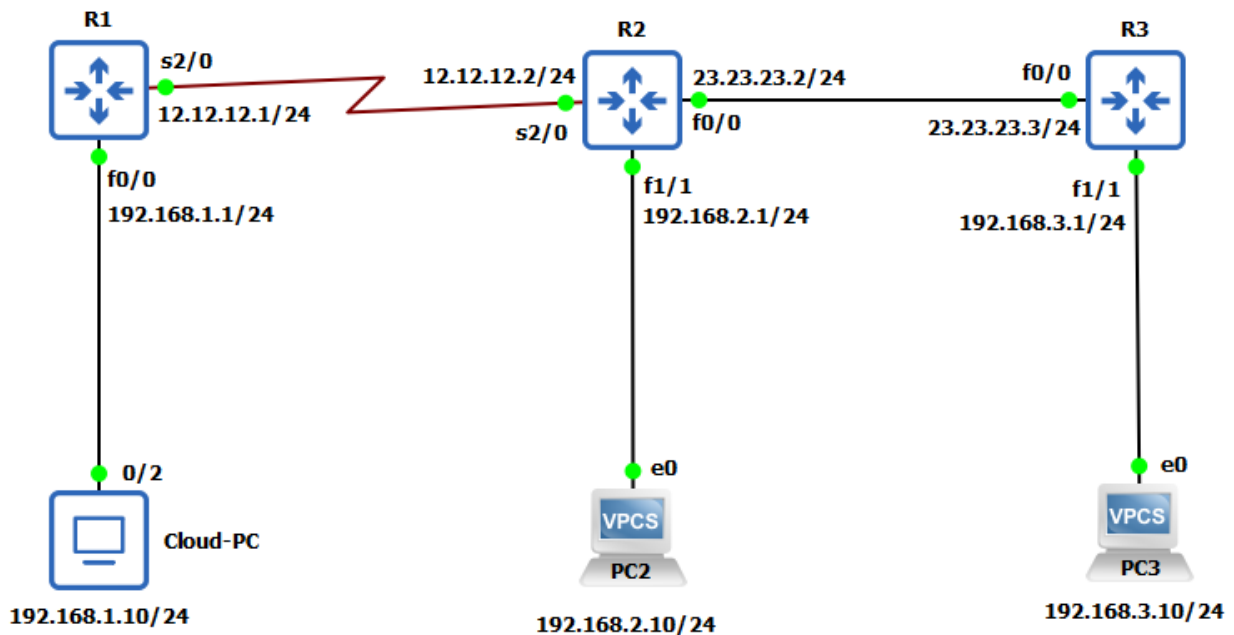


Router Administrative Management



Task

1. Create 2 users – User1 & User2 with password as Pass@1 & Pass@2
2. Encrypt clear text passwords and create MD5 encryption for enable password.
3. Create policy for minimum length password of 8 character.
4. Create policy to block the user for 120 sec when 2 login attempt fails within 30 sec.
5. Enable SSH for remote access.
6. Set clock and time zone

Note: All tasks are performed on R1 router.

Note: Cloud-PC is a cloud which bridge the R1 router with loopback adapter on real PC.

➤ User profile creation on R1.

```
R1#config t
R1(config)#username User1 password Pass@1
R1(config)#username User2 secret Pass@2
R1(config)#line vty 0 15
R1(config-line)#login local
R1(config-line)#exit
R1(config)#exit
R1#
```

✓ **Verification & Testing:****R1#sh run | section username**

username User1 password 0 Pass@1

username User2 secret 4 za6K1NJtAnDCceOm1WNVhTXfdlPyPcwOGuxn2Lc0cdY

R1#

➤ **Password policy creation on R1:**

R1#config t

R1(config)#enable secret ccna

The enable secret you have chosen is the same as your enable password.**This is not recommended. Re-enter the enable secret.**

R1(config)#enable secret ccnp

R1(config)#service password-encryption

R1(config)#security passwords min-length 8

R1(config)#username User3 password Pass@3

% Invalid Password length - must contain 8 to 25 characters. Password configuration failed

R1(config)#username User3 password Pass@123

R1(config)#exit

R1#

✓ **Verification & Testing:****R1#sh run | section username**

username User1 password 7 08114D5D1A3954

username User2 secret 4 za6K1NJtAnDCceOm1WNVhTXfdlPyPcwOGuxn2Lc0cdY

username User3 password 7 0236054818265E731F

R1#sh run | section enable

enable secret 4 Q1vGeU1vLYlZAzNHRI8P3ApXjj.YXC5QhC7CCyg0Izc

enable password 7 0007100805

Note: Number 7 is for cisco type 7 encryption, 5 is for MD5, 4 is for SHA256 and 0 is for clear text encryption.**R1#sh run**

Building configuration...

Current configuration : 2053 bytes

!

! Last configuration change at 18:40:48 UTC Mon Oct 5 2020

upgrade fpd auto

version 15.2

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption

!

hostname R1

!

```
boot-start-marker
boot-end-marker
!
!
security passwords min-length 8
enable secret 4 Q1vGeU1vLYlZAzNHRI8P3ApXjj.YXC5QhC7CCyg0Izc
enable password 7 0007100805
!
no aaa new-model
no ip icmp rate-limit unreachable
!
!
no ip domain lookup
ip cef
no ipv6 cef
!
multilink bundle-name authenticated
!
!
username User1 password 7 08114D5D1A3954
username User2 secret 4 za6K1NJtAnDCceOm1WNVhTXfdlPyPcwOGuxn2Lc0cdY
username User3 password 7 0236054818265E731F
!
redundancy
!
ip tcp synwait-time 5
!
interface FastEthernet0/0
description Link to local LAN
ip address 192.168.1.1 255.255.255.0
duplex auto
speed auto
!
interface FastEthernet0/1
no ip address
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0
no ip address
shutdown
duplex auto
speed auto
!
interface FastEthernet1/1
no ip address
shutdown
duplex auto
speed auto
!
```

```
interface Serial2/0
description Link to R2 on s2/0
ip address 12.12.12.1 255.255.255.0
serial restart-delay 0
clock rate 64000
!
interface Serial2/1
no ip address
shutdown
serial restart-delay 0
!
interface Serial2/2
no ip address
shutdown
serial restart-delay 0
!
interface Serial2/3
no ip address
shutdown
serial restart-delay 0
!
ip forward-protocol nd
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
control-plane
!
mgcp profile default
!
gatekeeper
shutdown
!
line con 0
exec-timeout 0 0
privilege level 15
password 7 070C285F4D0648
logging synchronous
login
stopbits 1
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
line vty 0 4
exec-timeout 1 0
password 7 070C285F4D06
logging synchronous
login local
```

```
transport input all
line vty 5 15
exec-timeout 1 0
password 7 070C285F4D06
logging synchronous
login local
transport input all
!
!
end
```

R1#

R1#sh run | section line

```
line con 0
exec-timeout 0 0
privilege level 15
password 7 070C285F4D0648
logging synchronous
login
stopbits 1
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
line vty 0 4
exec-timeout 1 0
password 7 070C285F4D06
logging synchronous
login local
transport input all
line vty 5 15
exec-timeout 1 0
password 7 070C285F4D06
logging synchronous
login local
transport input all
```

➤ **Login policy, SSH and clock configuration on R1:**

```
R1#config t
R1(config)#login block-for 120 attempts 2 within 30
R1(config)#login on-success log
R1(config)#login on-failure log
R1(config)#ip domain-name rsatechforum.in
R1(config)#crypto key generate rsa modulus 1024
```

The name for the keys will be: R1.rsatechforum.in

% The key modulus size is 1024 bits

% Generating 1024 bit RSA keys, keys will be non-exportable...

[OK] (elapsed time was 7 seconds)

```
R1(config)#line vty 0 15
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#exit
R1#
```

```
*Oct 1 18:52:40.007: %SSH-5-ENABLED: SSH 1.99 has been enabled
```

```
R1#clock set 19:30:00 1 Oct 2020
R1#config t
R1(config)#clock timezone IST +5 30
R1(config)#exit
R1#
```

✓ Verification & Testing:

```
R1#sh clock
```

```
19:31:07.847 IST Thu Oct 1 2020
R1#
```

```
R1#sh run | section line
```

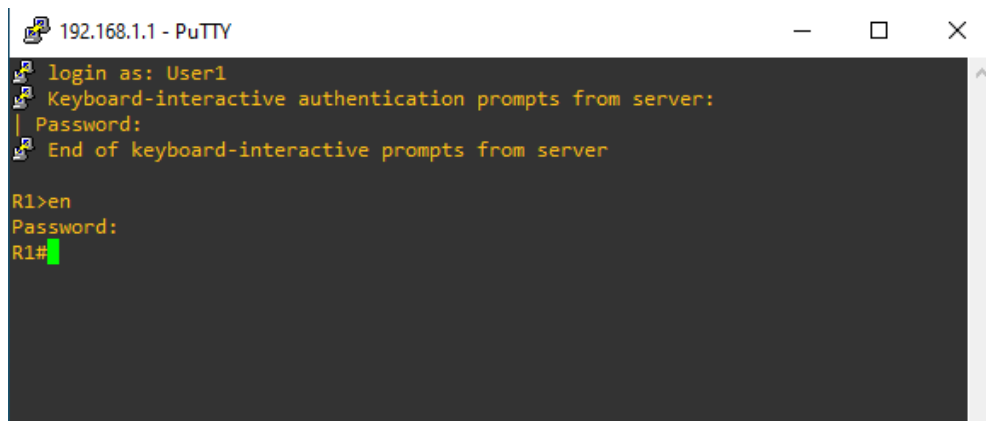
```
line con 0
exec-timeout 0 0
privilege level 15
password 7 070C285F4D0648
logging synchronous
login
stopbits 1
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
line vty 0 4
exec-timeout 1 0
password 7 070C285F4D06
logging synchronous
login local
transport input ssh
line vty 5 15
exec-timeout 1 0
password 7 070C285F4D06
logging synchronous
login local
transport input ssh
R1#
```

```
R1#sh ip ssh
```

```
SSH Enabled - version 1.99
```

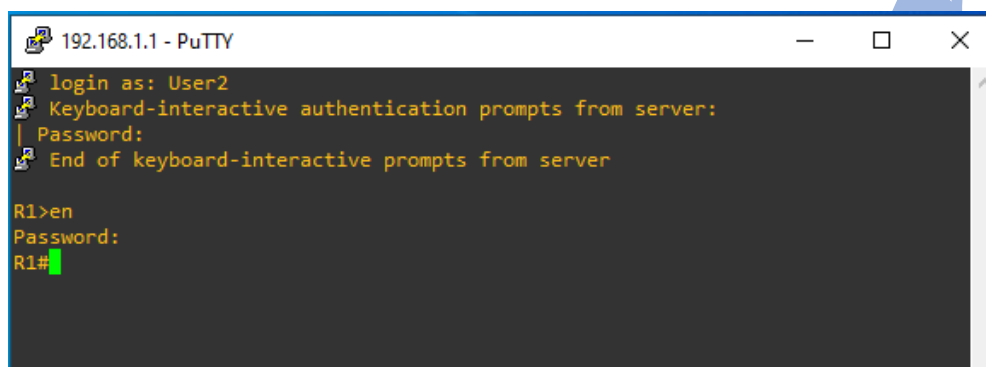
```
Authentication timeout: 120 secs; Authentication retries: 3
```

```
Minimum expected Diffie Hellman key size: 1024 bits
```

SSH to router R1 using Putty from Cloud-PC

```
192.168.1.1 - PuTTY
login as: User1
Keyboard-interactive authentication prompts from server:
Password:
End of keyboard-interactive prompts from server

R1>en
Password:
R1#
```



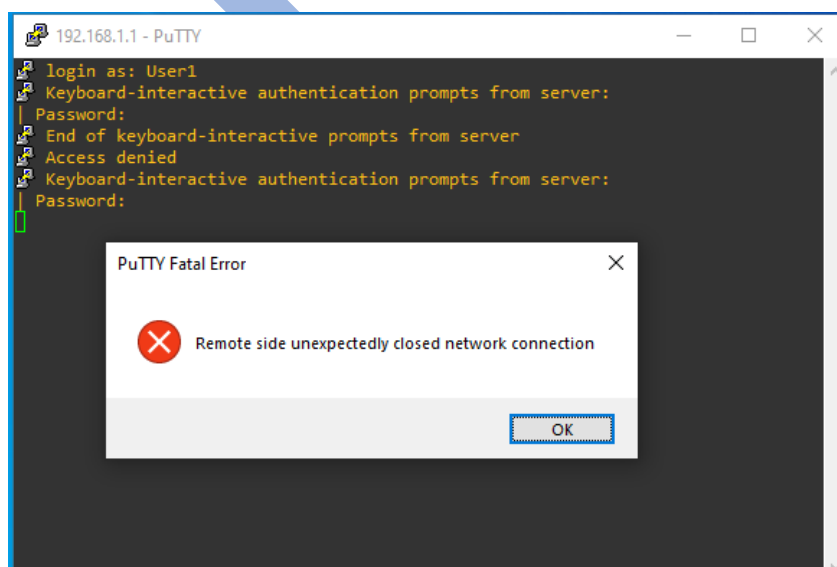
```
192.168.1.1 - PuTTY
login as: User2
Keyboard-interactive authentication prompts from server:
Password:
End of keyboard-interactive prompts from server

R1>en
Password:
R1#
```

R1#sh users

Line	User	Host(s)	Idle	Location
* 0 con 0		idle	00:00:00	
2 vty 0	User1	idle	00:00:48	192.168.1.10
3 vty 1	User2	idle	00:00:22	192.168.1.10

R1#

Make 2 login failure attempt from Cloud-PC

```
192.168.1.1 - PuTTY
login as: User1
Keyboard-interactive authentication prompts from server:
Password:
End of keyboard-interactive prompts from server
Access denied
Keyboard-interactive authentication prompts from server:
Password:
```

PuTTY Fatal Error

Remote side unexpectedly closed network connection

OK

R1#

*Oct 1 19:08:36.011: %SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: User1] [Source: 192.168.1.10] [localport: 22] [Reason: Login Authentication Failed] at 19:08:36 UTC Mon Oct 1 2020

R1#

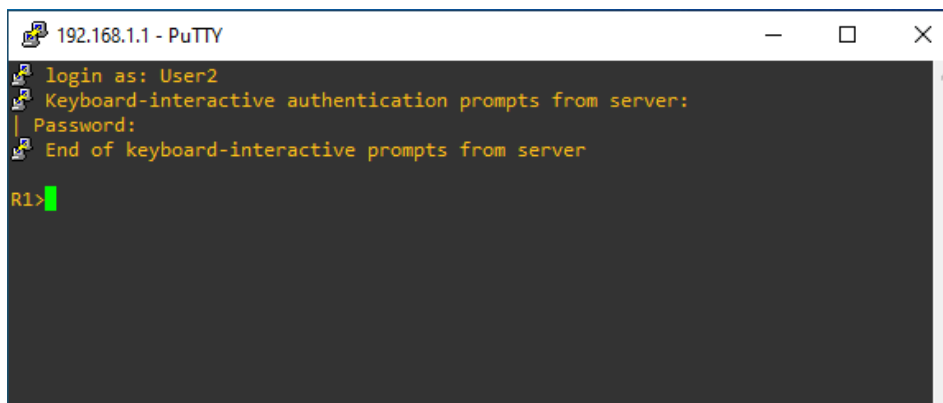
*Oct 1 19:08:42.027: %SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: User1] [Source: 192.168.1.10] [localport: 22] [Reason: Login Authentication Failed] at 19:08:42 UTC Mon Oct 1 2020

R1#

*Oct 1 19:08:42.031: %SEC_LOGIN-1-QUIET_MODE_ON: Still timeleft for watching failures is 18 secs, [user: User1] [Source: 192.168.1.10] [localport: 22] [Reason: Login Authentication Failed] [ACL: sl_def_acl] at 19:08:42 UTC Mon Oct 1 2020

R1#

Reattempt to login with correct credential after 120 sec



R1#

*Oct 1 19:10:42.031: %SEC_LOGIN-5-QUIET_MODE_OFF: Quiet Mode is OFF, because block period timed out at 19:10:42 UTC Mon Oct 1 2020

R1#

*Oct 1 19:10:59.751: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: User2] [Source: 192.168.1.10] [localport: 22] at 19:10:59 UTC Mon Oct 1 2020

R1#

R1#sh login

A default login delay of 1 seconds is applied.
No Quiet-Mode access list has been configured.
All successful login is logged.
All failed login is logged.

Router enabled to watch for login Attacks.
If more than 2 login failures occur in 30 seconds or less,
logins will be disabled for 120 seconds.

Router presently in Normal-Mode.
Current Watch Window
Time remaining: 2 seconds.
Login failures for current window: 0.
Total login failures: 2.

R1#sh login failures

Total failed logins: 2

Detailed information about last 50 failures

Username	SourceIPAddr	lPort	Count	TimeStamp
User1	192.168.1.10	22	2	19:08:42 UTC Mon Oct 1 2020

R1#

Ratan

Important Commands:

sh run | section username
sh run | section line
sh clock
sh ip ssh
sh users
sh login
sh login failures

Ratan